Advance subnetting & VLSM

1. Given a company with 3 departments (HR: 20 users, Sales: 50 users, Tech: 100 users), assign appropriate subnets using VLSM from the 192.168.10.0/24 network and write the subnet address, subnet mask, and usable IP range for each department.

ANS:

VLSM Subnetting — 192.168.10.0/24

Using **VLSM**, we allocate the largest department first.

1. Tech — 100 users

Needs at least 100 usable IPs.

- /25 → 128 total, **126 usable**
- Network: 192.168.10.0/25
- Mask: 255.255.255.128
- Usable: 192.168.10.1 – 192.168.10.126

2. Sales — 50 users

Needs at least 50 usable IPs.

- /26 → 64 total, **62 usable**
- Network: 192.168.10.128/26
- Mask: 255.255.255.192
- Usable: 192.168.10.129 – 192.168.10.190

3. HR — 20 users

Needs at least 20 usable IPs.

- /27 → 32 total, **30 usable**
- Network: 192.168.10.192/27
- Mask: 255.255.255.224

- Usable: 192.168.10.193 – 192.168.10.222

Final Table

Department	Users	Subnet	Subnet Mask	Usable IP Range
Tech	100	192.168.10.0/25	255.255.255.128	192.168.10.1 – 192.168.10.126
Sales	50	192.168.10.128/26	255.255.255.192	192.168.10.129 – 192.168.10.190
HR	20	192.168.10.192/27	255.255.255.224	192.168.10.193 – 192.168.10.222

2. You are given three office locations (Ahmedabad, Surat, Rajkot). Each office has three departments: Admin (10 users), Marketing (30 users), and Support (60 users). Design an IP addressing scheme using VLSM for all offices using the 10.0.0.0/22 block, ensuring no overlap and efficient usage. List each subnet with its office, department, subnet mask, and first/last usable IP.

ANS:

LSM IP Addressing Scheme — 10.0.0.0/22

Each office needs:

- **Support: 60 users → /26** = 62 usable IPs
- **Marketing: 30 users → /27** = 30 usable IPs
- **Admin: 10 users → /28** = 14 usable IPs

We allocate the **largest subnet first** and then continue sequentially, ensuring there is **no overlap**.

Final VLSM Table

Office	Department	Subnet	Subnet Mask	First Usable IP	Last Usable IP
Ahmedabad	Support	10.0.0.0/26	255.255.255.192	10.0.0.1	10.0.0.62
Ahmedabad	Marketing	10.0.0.64/27	255.255.255.224	10.0.0.65	10.0.0.94
Ahmedabad	Admin	10.0.0.96/28	255.255.255.240	10.0.0.97	10.0.0.110
Surat	Support	10.0.0.128/26	255.255.255.192	10.0.0.129	10.0.0.190
Surat	Marketing	10.0.0.192/27	255.255.255.224	10.0.0.193	10.0.0.222
Surat	Admin	10.0.0.224/28	255.255.255.240	10.0.0.225	10.0.0.238
Rajkot	Support	10.0.1.0/26	255.255.255.192	10.0.1.1	10.0.1.62
Rajkot	Marketing	10.0.1.64/27	255.255.255.224	10.0.1.65	10.0.1.94
Rajkot	Admin	10.0.1.96/28	255.255.255.240	10.0.1.97	10.0.1.110

How the allocation works

Ahmedabad:
/26 → /27 → /28

Surat:
/26 → /27 → /28

Rajkot:
/26 → /27 → /28

The remaining portion of the 10.0.0.0/22 block is unused and available for **future departments or expansion**.

**3. A startup has offices in Mumbai, Pune, and Baroda. Each office will connect to HQ via VPN. Propose a route summarization for all three offices if their assigned subnets are: Mumbai (172.16.10.0/24), Pune (172.16.11.0/24), Baroda (172.16.12.0/24). Write the summarized route and explain why it is efficient.

Hint: Find the smallest supernet that includes all three subnets.**

ANS:

Route Summarization

Given office subnets:

Office	**Subnet**
Mumbai	172.16.10.0/24
Pune	172.16.11.0/24
Baroda	172.16.12.0/24

The smallest supernet that includes all three is:

Summarized Route: 172.16.8.0/21

- Subnet mask: **255.255.248.0**
- Covers: **172.16.8.0 – 172.16.15.255**
- Therefore, it includes:
    - 172.16.10.0/24
    - 172.16.11.0/24
    - 172.16.12.0/24

Why is it efficient?

Instead of HQ maintaining **3 separate routes**, it can maintain **one summarized route**:

172.16.8.0/21

This reduces the routing table size and makes routing simpler and more efficient.

Answer:

Summarized route = 172.16.8.0/21
It covers all three office networks with a single routing entry.

**4. Given a scenario where the Tech department in Ahmedabad grows from 100 to 200 users, update your previous VLSM plan to accommodate the new size with minimal changes to the rest of the network. List the new subnet details and explain your approach.

Constraint: Do not change the existing subnets for HR or Sales unless absolutely necessary.**

ANS:

Updated VLSM Plan – Ahmedabad Tech Department

Since the **Tech department grows from 100 to 200 users**, we need a larger subnet.

For **200 users**:

- /25 → 126 usable IPs ❌
- /24 → 254 usable IPs ✅

So, Tech needs a **/24 subnet**.

To keep changes to HR and Sales minimal, we can move only the Tech subnet into unused address space.

New Subnet Details

Department	New Subnet	Subnet Mask	Usable IP Range	Usable Hosts
Ahmedabad Tech	**10.0.1.0/24**	255.255.255.0	10.0.1.1 – 10.0.1.254	254

Approach

1. **HR and Sales remain unchanged** as required.
2. Ahmedabad Tech is increased to a **/24** subnet because 200 users need at least 254 usable addresses.
3. The new Tech subnet is placed in **unused address space** (10.0.1.0/24).
4. This avoids overlapping with the existing HR and Sales networks.
5. The change is therefore limited mainly to the **Tech department and its routing/DHCP configuration**.

Final: Ahmedabad Tech → **10.0.1.0/24**, supporting **254 usable hosts**.

6. **Create 5 VLSM subnetting challenge questions inspired by real apps you use (e.g., Zomato delivery zones, Flipkart warehouse locations, or IPL team fan clubs). For each, provide the required number of hosts and a base network address.**

ANS:

5 VLSM subnetting challenge questions based on real-world app scenarios. These are designed for you to **solve yourself**—I have not included the answers.

1. Zomato Delivery Zones 🍔

Zomato wants to divide a city into delivery zones.

- **Base Network:** 192.168.10.0/24
- **Zone A:** 60 delivery devices
- **Zone B:** 30 delivery devices
- **Zone C:** 12 delivery devices

Challenge: Use VLSM to create suitable subnets for all three zones.

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2. Flipkart Warehouse Network 📦

A Flipkart warehouse needs separate networks for different departments.

- **Base Network:** 10.10.0.0/24
- **Packing:** 100 hosts
- **Inventory:** 50 hosts
- **Management:** 20 hosts

Challenge: Allocate VLSM subnets efficiently with no overlapping addresses.

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3. IPL Team Fan Club 🏏

An IPL team creates separate networks for its online fan-club events.

- **Base Network:** 172.16.0.0/24
- **Main Fan Zone:** 120 hosts
- **VIP Fan Zone:** 50 hosts
- **Staff Zone:** 20 hosts

Challenge: Apply VLSM and find the subnet address, prefix, usable host range, and broadcast address for each zone.

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4. Swiggy Delivery Hub 🛵

A Swiggy delivery hub needs networks for its delivery operations.

- **Base Network:** 192.168.20.0/24
- **Delivery Riders:** 80 hosts
- **Restaurant Partners:** 40 hosts
- **Hub Staff:** 15 hosts

Challenge: Design a VLSM plan using the smallest suitable subnet for each group.

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5. Instagram Creator Office 📱

An Instagram creator-management office needs separate networks.

- **Base Network:** 10.20.0.0/24
- **Content Team:** 70 hosts
- **Marketing Team:** 35 hosts

- **Admin Team:** 10 hosts

Challenge: Use VLSM to allocate addresses efficiently and provide the subnet mask, usable host range, and broadcast address for each team.